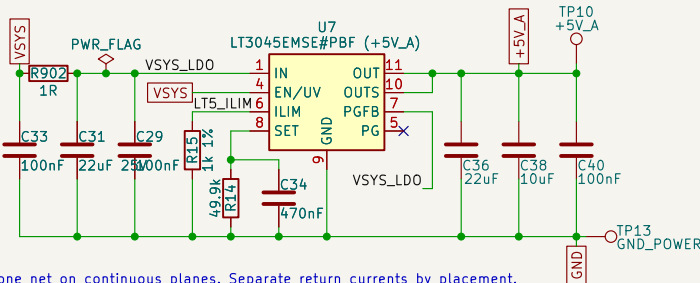




Battery and power

SYSTEM RAIL VSYS (from BQ25798 SYS, Battery/charger/USB-C sheet)

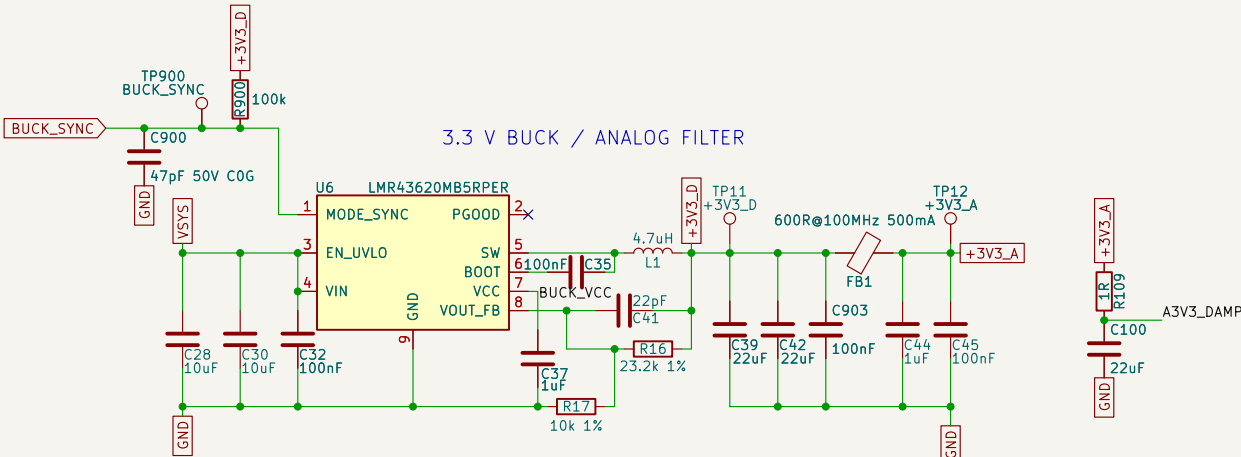
+5 V ANALOG LDO



GND is one net on continuous planes. Separate return currents by placement.
FB1 filters the positive supply; it is not a ground connection.

U6: adjustable output, forced PWM synchronised to BUCK_SYNC (833.33 kHz = ADC fs). C34 sets the LDO noise tradeoff.
Connect L1 start lead to SW. Its magnetic core requires field qualification.

3.3 V BUCK / ANALOG FILTER



U6: R16/R17 23.2k / 10k set 3.320 V nominal (SNVSBY5B Eq. 5; RFBT|IRFBB 6.99k). C41 spans the upper resistor.
Keep the feedback node away from SW. L1 XGL4020-472MEC: Isat 3.0 A (Coilcraft Doc 1529-1).

Module interfaces

Signal / bus	I/O	Source / destination	Purpose
VSYS from U301	IN	BQ25798 SYS (U301) protected VSYS; the BQ25798 charger is on the 'Battery, charger and USB-C' sheet.	Common GND net; separate analog and digital placement
VSYS	PWR OUT VSYS -> U6, U7, MCU sense, backlight, EXPSYS	System rail distribution and VSYS monitoring.	Prototype: bench validation required
+3V3_D	PWR OUT	U6/L1 -> digital sheets	Buck supply for MCU, SD and PSRAM.
+3V3_A	PWR OUT	+3V3_D through FB1 -> Analog/MCU	Filtered positive analog supply.
+5V_A	PWR OUT	U7 -> RF/Analog/MCU	Low-noise detector/reference supply.
SW / BOOT / BUCK_VCC	LOCAL	U6,L1,C35,C37	Switching and regulator bypass loops.
FB / SET / ILIM / OUTS	LOCAL	U6 divider / U7 control pins	Regulation, limit and output sensing.
GND	PWR	All return pins -> GND	Separate current paths by placement.

Common GND net; separate analog and digital placement
Prototype: bench validation required

Senior Design / RFVD

Sheet: /Battery and power/
File: Power.kicad_sch

Title: Battery and power

Size: USLetter Date: 2026-09-07

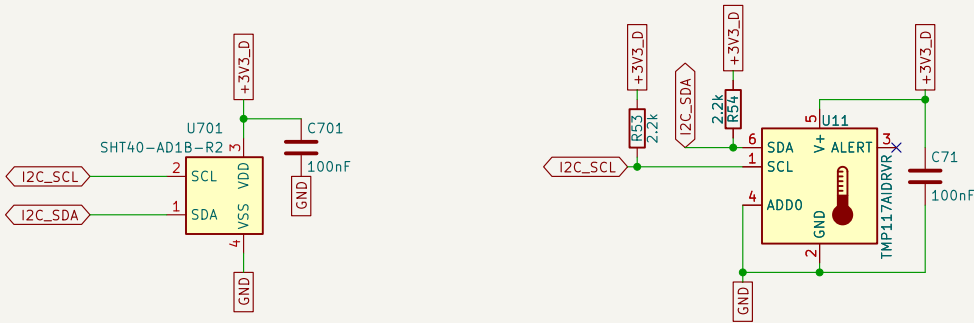
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Rev: SENSOR-D

Id: 2/12

Signal / bus	I/O	Source / destination	Purpose
I2C_SCL / I2C_SDA	I/O	MCU <-> U11	Open-drain temperature register bus.
ADD0	LOCAL	U11 pin 4 -> GND	Fixed I2C address strap.
ALERT	LOCAL	U11 pin 3 unused	Polling is used; no alert connection.
+3V3_D / C71 / GND	PWR	Power -> U11	Board temperature and local bypass.

Board temperature monitor



ADD0 = GND; ALERT unused
EP intentionally floating and unsoldered; use project NoPasteEP footprint.
Temperature is a calibration-condition observation, not implant temperature.

Common GND net; separate analog and digital placement
Prototype: bench validation required

Senior Design / RFVD

Sheet: /Temperature monitor/
File: Temperature.kicad_sch

Title: Board temperature monitor

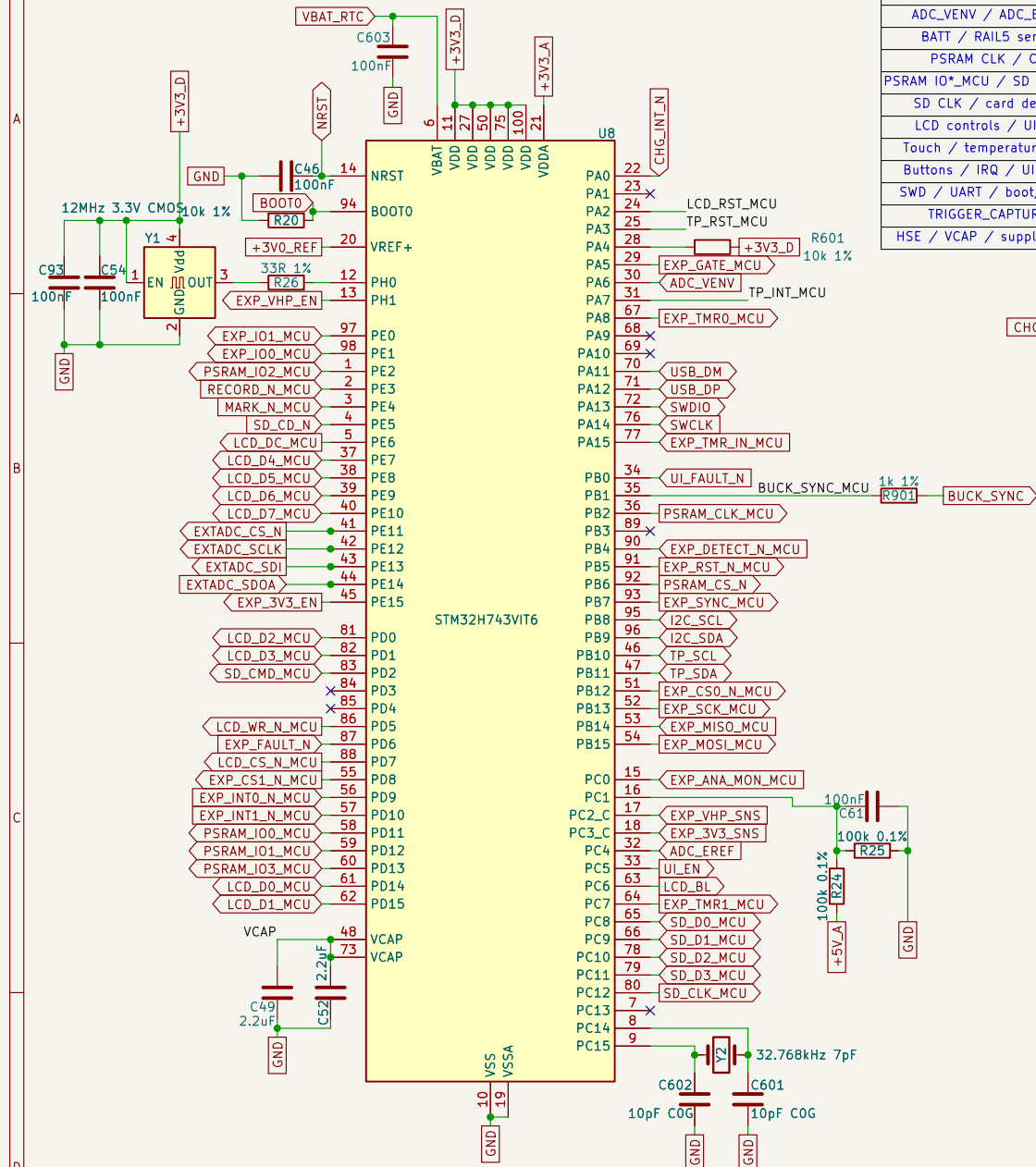
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Rev: SENSOR-D
Id: 3/12

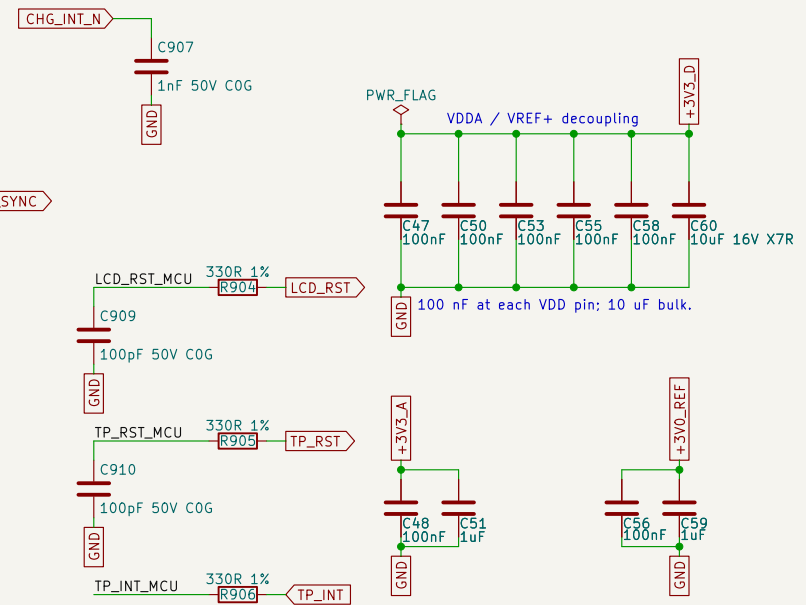
STM32H743 controller

Module interfaces

Signal / bus	I/O	Source / destination	Purpose
ADC_VENV / ADC_EREf	IN	Analog -> PA6 / PC4	Simultaneous envelope/reference pair.
BATT / RAIL5 sense	LOCAL	R24/R25 -> ADC3 (+5V_A)	Slow supply-health observations.
PSRAM CLK / CE	OUT	MCU -> Storage	Memory clock and selection.
PSRAM IO*_MCU / SD CMD_DAT	I/O	MCU <-> Storage	Buffered capture and card transfers.
SD CLK / card detect	OUT / IN	MCU <-> Storage	Card timing and insertion state.
LCD controls / UL_EN	OUT	MCU -> Display	Screen data, reset, backlight and power.
Touch / temperature I2C	I/O	MCU <-> Display/Temperature	Open-drain control and sensor buses.
Buttons / IRQ / UI fault	IN	Display -> MCU	User actions and module status.
SWD / UART / boot/reset	I/O	MCU <-> Service	Programming, diagnosis and recovery.
TRIGGER_CAPTURE	IN	Trigger -> PA0	Optional edge timestamp; not ADC clock.
HSE / VCAP / supply rails	LOCAL / PWR	Y1,R26,C49,C52 and rails	Clock, core bypass and common return.



12 MHz HSE bypass (PLL_M = 6); PH1 = EXP_VHP_EN (GPIO, OSC_OUT unused in bypass). PB1 = TIM1_CH3N -> BUCK_SYNC.
 PA6 and PC4 form the simultaneous ADC pair. SWD is the programming interface.
 ADC3 monitors EXP_ANA_MON (PC0) and analog 5 V (2:1 divider, PC1).
 VBAT/VSYS: U301 BQ25798 ADC over I2C1. Apply rail-validity and startup guards.



C908 (DNP RC footprint on TP_INT_MCU) deleted (S-9); R906 330R stays the only TP_INT conditioning.
 PA7 is driven low (output) during RECORD. PB4 (pin 85A) unused. Display bus is active only (external loads).

S-16/18/24, SC-11 (2026-10-04): SW4/R801 removed, BOOT0 = R20 only (DFU via FW stub).
 PB3 (ex-SWD), PD3 (ex-LID_INT), PA1 (ex-LCD_TE) not connected, set analog. OPT3004 removed.
 PE3/PE4 (behind 1k R46/R47) and NRST: internal pull-ups; R18, R44, R49, SW3 removed.

Common GND net; separate analog and digital placement
 Prototype: bench validation required

Senior Design / RFVD

Sheet: /STM32H743 controller/
 File: MCU.kicad_sch

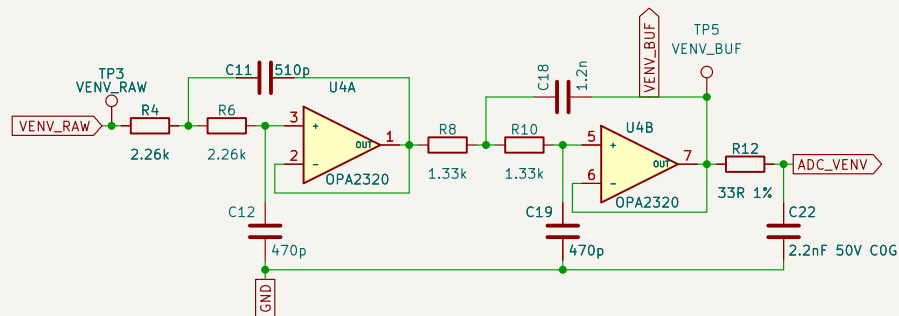
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Size: USLetter Date: 2026-09-07
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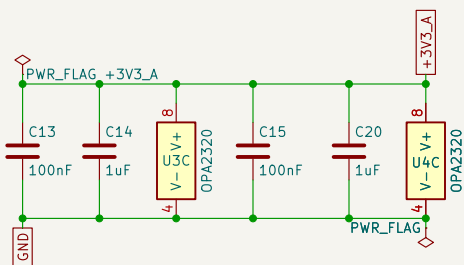
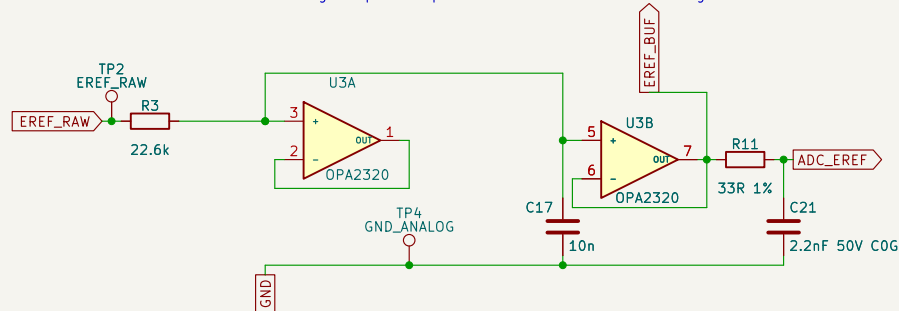
Rev: SENSOR-D
 Id: 4/12

Envelope and reference filters

About 95 kHz modeled envelope bandwidth. Apply digital filtering before decimation.
Qualify settling, clipping and noise on the assembled board.



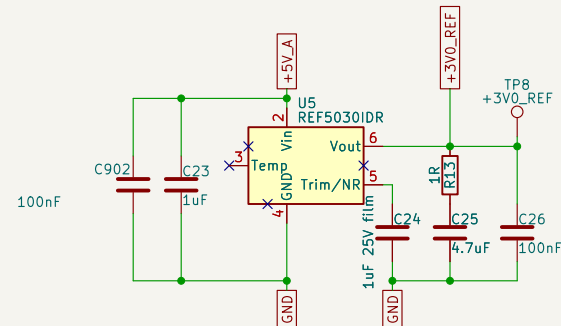
TP2/TP3: raw pair. ADC pair: probe C22 / C21 pad 1 (TP7/TP6 removed). TP5: envelope buffer.
Use high-impedance probes and account for their loading.



100 nF + 1 uF at each dual amplifier supply. Both packages use +3V3_A.

Module interfaces

Signal / bus	I/O	Source / destination	Purpose
VENV_RAW / EREF_RAW	IN	RF input -> R4 / R3	Envelope and reference filter inputs.
ADC_VENV / ADC_EREf	OUT	R12 / R11 -> MCU	Filtered pair; probe C22 / C21 pad 1.
U4 / U3 filter loops	LOCAL	Each output -> its minus input	Unity-gain Sallen-Key feedback.
VENV_BUF / TP5	LOCAL	U4 pin 7 -> R12	Envelope output before ADC isolation.
C22 / C21	LOCAL	ADC inputs -> GND	Local charge reservoirs after 47R.
+3V0_REF	PWR OUT	U5 -> MCU VREF+	Shared conversion reference.
+5V_A / +3V3_A	PWR IN	Power -> U5 / U3,U4	Reference and amplifier supplies.
GND	PWR	Common ground plane	Keep bypass and feedback returns local.



Common GND net; separate analog and digital placement
Prototype: bench validation required

Senior Design / RFVD

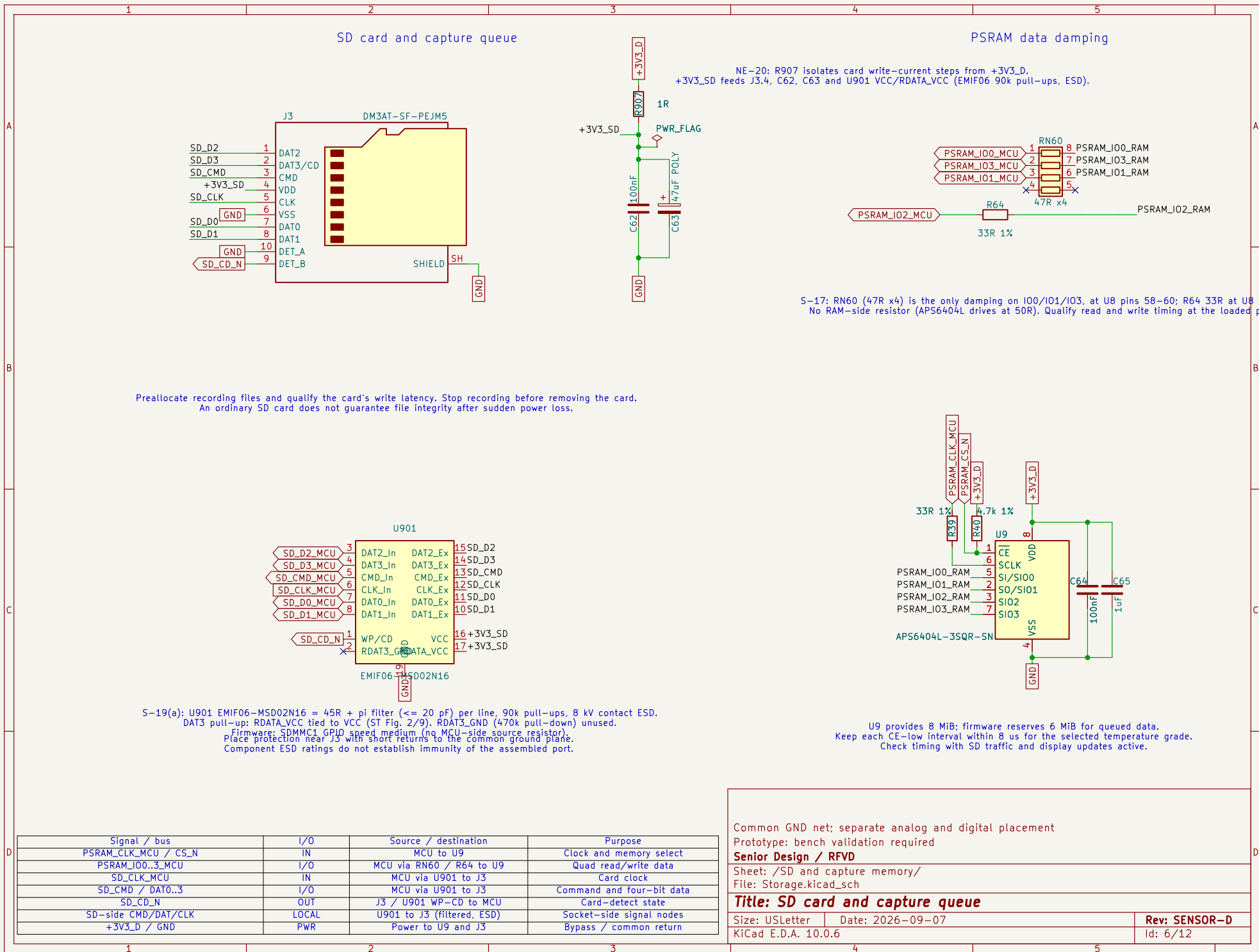
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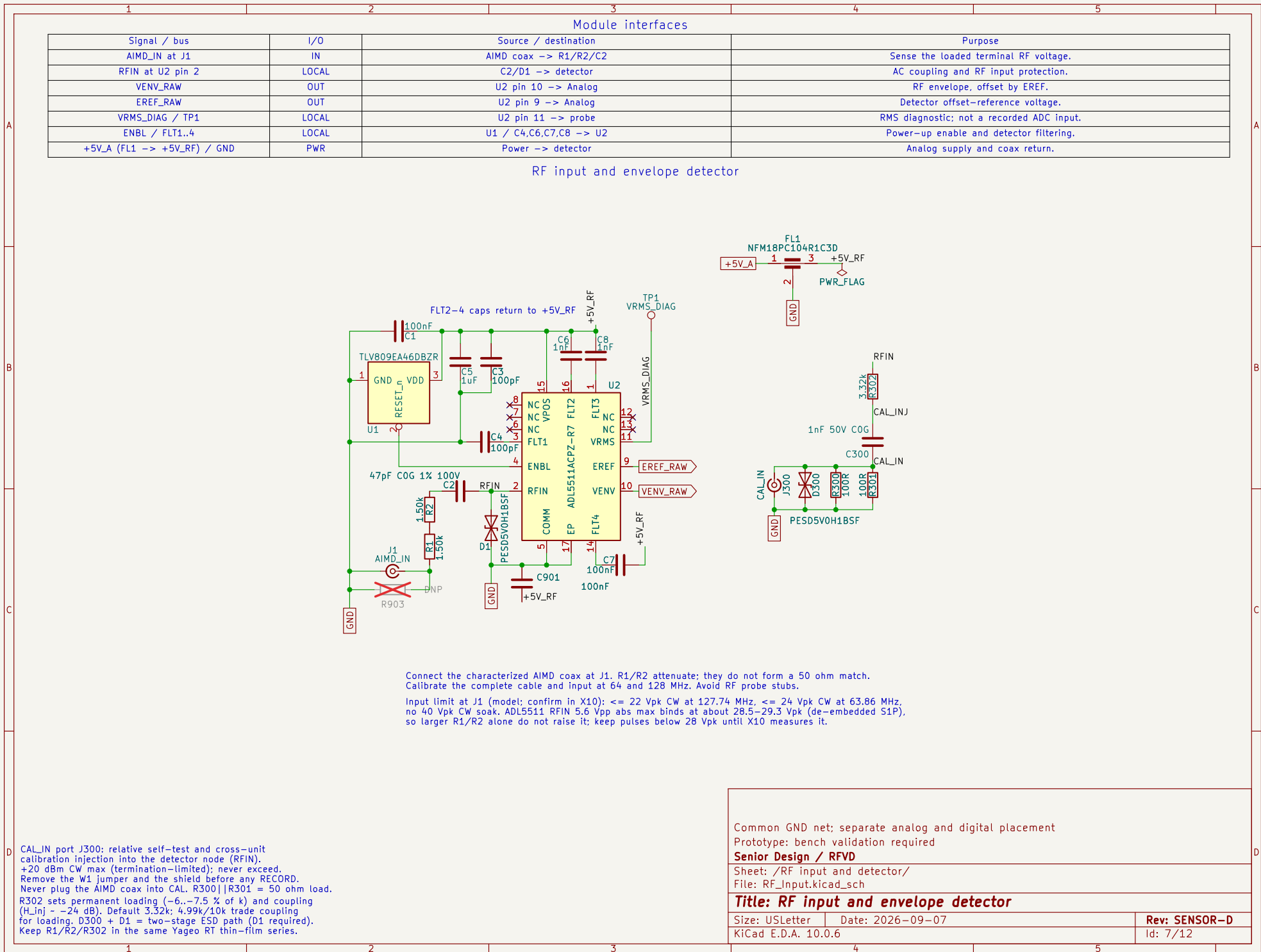
Title: Envelope and reference filters

Size: USLetter Date: 2026-09-07
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Rev: SENSOR-D
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Filter resistors: 0.1%; filter capacitors: COG 1% (C17 EREF pole and C21/C22 ADC reservoirs: COG 5%). ADC range: AD7380 0 to 2.5 V (internal reference: 3.2V). BB ADC 0 to +3V0_REF
Amplifier overdrive can clip the ADC; check both channels during calibration.

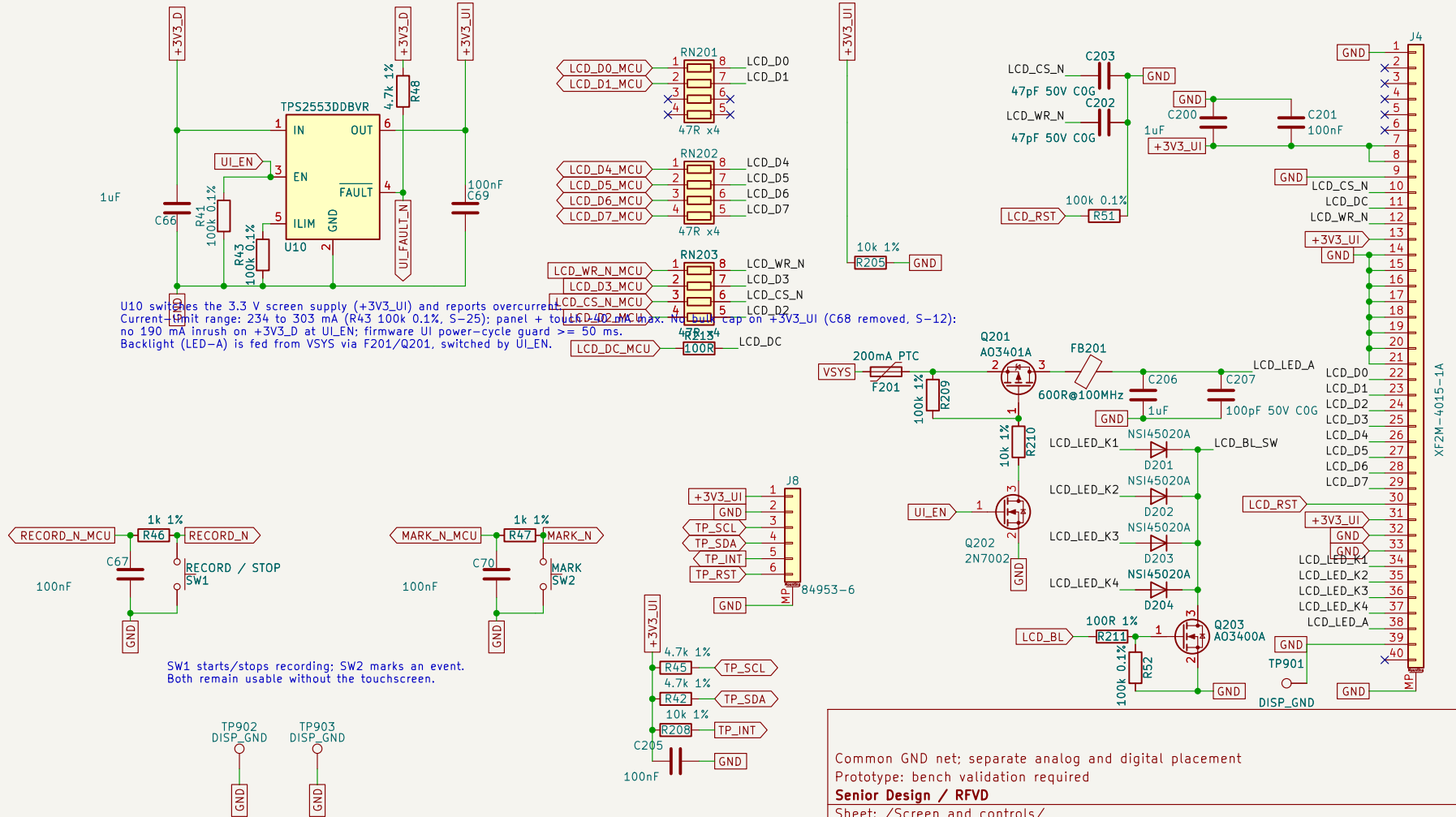




Display, touch and physical controls

Signal / bus	I/O	Source / destination	Purpose
LCD FMC D0–D7/WR/RD/CS/DC, RST, BL	IN	MCU to J4	Display data and control
TP_SCL / TP_SDA	I/O	MCU and touch module	Touch I2C; +3V3_UI pull-ups
TP_RST / TP_INT	IN / OUT	MCU and touch module	Reset / touch-ready indication
UI_EN / UI_FAULT_N	IN / OUT	MCU and U10	Power enable / overcurrent flag
RECORD_N / MARK_N	OUT	SW1 / SW2 via 1k R46 / R47 to MCU	Active-low recording controls
+3V3_UI / local bypass	LOCAL	U10 to J4	Switched screen rail; sequence I/O
+3V3_D / GND	PWR	Power sheet / common return	

J4: Newhaven NHD-2.4-240320AF-CSXP-CTP (ST7580) 800x480 pixels, 60 Hz, write-only) via Omron XF2M-4015-1A; also fits NHD-2.8-...-FCTP, IM2; IM0 = 001 hard-tied (NHD Interface Selection p.5); RDX (J4.13) tied to +3V3_UI, PD4 (FMC_NOE) unused; firmware never reads the panel (S-11); J8: FT5426 touch (I2C2, 0x38), FMC bank 1 NE1, D/C = A22; TE (J4.40) not connected (FEAT-6). Display and backlight are unpowered (UI_EN low) during acquisition.

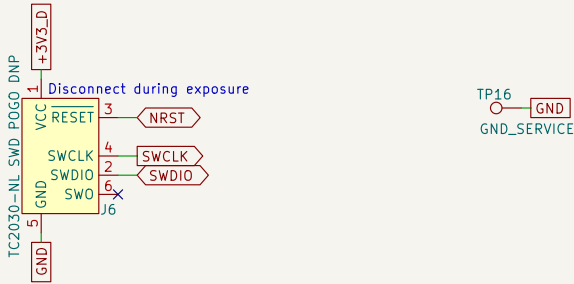


Start with UI power off; set safe GPIO states before switching power.
 Acquisition and SD service take priority over rendering. Calibration needs a known external stimulus.

Module interfaces			
Signal / bus	I/O	Source / destination	Purpose
SWCLK / SWDIO (SW0 NC)	OUT / I/O	J6 probe <=> MCU	Programming; trace via RTT/USB CDC.
PA9 / PA10 (ex-UART1)	NC	none (J7 removed)	Spare; console on USB CDC.
NRST	I/O	J6 <=> MCU	External and MCU reset path.
BOOT0	LOCAL	R20 pad 1 at U8 pin 94	Bench recovery: strap to +3V3_D.
DFU entry	FW	RECORD+MARK chord / USB	Boot stub jumps to ROM DFU.
+3V3_D / GND	PWR	Board -> probe voltage sense	3.3V logic reference; avoid backpower.

Programming and service

SWD POGO / SERVICE ONLY



Common GND net; separate analog and digital placement
Prototype: bench validation required
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Sheet: /Programming and service/
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Title: Programming and service

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